Temporal Envelopes from Complex-Base Dynamics: Möbius Collection → Quench → Fractal Circular/Spiral Growth → Re-Closure

VR & AI

August 24, 2025

Abstract

We present a self-contained dynamical theory of *temporal envelopes*—Möbius, circular, and spiral—built on a configurable complex-base formalism with parameters (s, b, l). The state is a hierarchical, phase-aware integer register over residue tracks modulo l, normalized by a local tri-site identity derived from a quadratic algebraic base b. A *tick* deposits potential energy into the register and normalization redistributes it. With Möbius holonomy (orientation reversal per loop), ticks *collect* without directional drift; upon a threshold (*quench*) the holonomy flips and the system enters a circular ($\kappa = 0$) or spiral ($\kappa \neq 0$) propagation regime that spawns *fractal child* circles/spirals across levels via the shrink factor s. A termination rule re-closes the envelope to Möbius. We give precise update maps, Lyapunov bounds, and invariance of the normalization calculus, and illustrate with the φ–$l = 4$ quarter-turn lattice, including a figure-8 half-twist macrocycle.

Contents

1 Primer: the (s, b, l) complex-base formalism **2**
1.1 Sites, tracks, and values . . . 2
1.2 Quadratic base and tri-site identity . . . 2

2 Action Principle for the Tri-Site Identity **2**

3 Why $(\varphi, l = 4)$ **3**

4 Assumptions, Thresholds, and Reproducible Rules **3**

5 Temporal envelopes and ticks **5**
5.1 State . . . 6
5.2 Tick operator and normalization . . . 6
5.3 Macrocycle and holonomy action . . . 6

6 Prediction: Chebyshev–Lucas Radial Law **6**

7 Stage I (Möbius): collection without drift **7**

8 Quench: unwrap to circular or spiral **7**

9 Stage II (circular/spiral): propagation and fractal children **7**
9.1 Fractal spawn across levels . . . 7

10 Termination and re-closure to Möbius **8**

11 Algorithms (macrocycle skeleton) 8

12 Worked example: φ with quarter-turn lattice 8

13 Discussion and outlook 8

14 Bridge to Curvature 9

1 Primer: the (s, b, l) complex-base formalism

We recall the essentials; see [1] for the full normalization theory and proofs.

1.1 Sites, tracks, and values

Fix $l \geq 2$ and $b > 1$. Let $U = e^{i2\pi/l}$ and $B = bU$. Indices $k \in \mathbb{Z}$ decompose into l *phase tracks* by residue $r \equiv k \pmod{l}$, where $k = r + lt$ and $t \in \mathbb{Z}$. On a fixed track r we write $d_t := d_{r+lt}$.

Definition 1.1 (Strings and hierarchical sums)**.** A (single-level) (b, l)-string is an integer array $(d_k)_{k\in\mathbb{Z}}$ with finite right tail and absolutely summable left tail; its value is $\mathrm{Val}(d) = \sum_{k\in\mathbb{Z}} d_k B^k$ (absolutely convergent). For multi-level hierarchies, fix a shrink $s \in (0, 1]$ and arrays $d_{\ell,k}$ $(\ell = 0, 1, \dots)$; the value is

$$Z \;=\; \sum_{\ell\geq 0} s^\ell \sum_{k\in\mathbb{Z}} d_{\ell,k}\, B^k. \tag{1}$$

1.2 Quadratic base and tri-site identity

Assume b solves $b^2 - Pb + Q = 0$ with $P, Q \in \mathbb{Z}$, $b > 1$. Let $V_n(P, Q)$ be the Lucas sequence. Then for $n = l$ one has $b^{2l} = V_l b^l - Q^l$ and the *phase-aligned identity*

$$B^{k+2l} - V_l\, B^{k+l} + Q^l\, B^k \;=\; 0, \qquad (k \equiv r \bmod l). \tag{2}$$

This induces a *tri-site* carry/borrow on each track and a deterministic, rightmost-first normalization that terminates and is confluent (see [1] for proofs).

Remark 1.2 (Lyapunov potential). Fix $q \in (1, \beta_l)$ with $\beta_l = b^l$. The weighted potential $E_q = \sum_t |d_t|\, q^t$ strictly decreases under a unit carry at site t, and hence normalization terminates.

2 Action Principle for the Tri-Site Identity

Fix a residue track and consider a complex field $\psi = \{\psi_t\}_{t\in\mathbb{Z}}$. Define the second-order difference operator

$$(\mathcal{L}\psi)_t := \psi_{t+1} - V_l\, \psi_t + Q^l\, \psi_{t-1}.$$

Proposition 2.1 (Constrained action)**.** *Let*

$$S[\psi, \lambda] \;=\; \sum_{t\in\mathbb{Z}} \lambda_t\, (\mathcal{L}\psi)_t,$$

with Lagrange multipliers $\lambda_t \in \mathbb{C}$. The Euler–Lagrange equations are

$$\frac{\partial S}{\partial \lambda_t} = 0 \;\Rightarrow\; (\mathcal{L}\psi)_t = 0, \qquad \frac{\partial S}{\partial \psi_t} = 0 \;\Rightarrow\; (\mathcal{L}^*\lambda)_t = 0,$$

so extremals lie in $\ker \mathcal{L}$.

Remark 2.2 (Unconstrained variants). (i) Penalized action: $S_\mu[\psi] = \frac{1}{2}\sum_t |(\mathcal{L}\psi)_t|^2 + \frac{\mu}{2}\sum_t |\psi_t|^2$. Global minimizers satisfy $\mathcal{L}\psi = 0$ as $\mu \downarrow 0$. (ii) Phase-locked harmonic action:

$$S_\gamma[\psi] = \tfrac{1}{2}\sum_t \left|\psi_{t+1} - b\, e^{i2\pi/l}\psi_t\right|^2 + \tfrac{\gamma}{2}\sum_t |\psi_t|^2.$$

Choosing γ so that the characteristic roots at l-spaced sites match the Lucas relation reproduces (2) in l-steps.

3 Why $(\varphi,\, l = 4)$

Proposition 3.1 (Minimal half-twist lattice)**.** *A two-block macrocycle with equal scales and one site-step per block has monodromy -1 iff $l = 4$.*

Proof. The macro angle is $2 \times \frac{2\pi}{l} = \pi$. Hence $l = 4$. □

Proposition 3.2 (Symmetric collection)**.** *For metallic means $(Q = -1)$, the condition $Q^l = +1$ needed for symmetric neighbor updates holds iff l is even; $l = 4$ is the minimal lattice compatible with the previous proposition.*

Proposition 3.3 (Economy among metallic bases)**.** *For fixed even l, both b^l and $V_l(P, -1)$ increase with $P \geq 1$. Hence $(P, Q) = (1, -1)$ (golden ratio) yields the smallest digit set and simplest carry at that l.*

4 Assumptions, Thresholds, and Reproducible Rules

Standing assumptions

- **Quadratic base and tri-site rule.** Fix $b > 1$ solving $b^2 - Pb + Q = 0$ and an even lattice $l \geq 2$. Let $U = e^{i2\pi/l}$, $B = bU$, $\beta_l = b^l > 1$, and $V_l = V_l(P, Q)$. The phase-aligned identity (2) induces tri-site carries per track with deterministic normalization (rightmost-first), which *terminates and is confluent* (see normalization theorem in the math draft).
- **Collection symmetry.** We assume $Q^l = +1$ throughout the Möbius collection stage so that neighbor updates are symmetric.
- **Digits.** Use the nonnegative digit set $\mathcal{D} = \{0, 1, \ldots, V_l - 1\}$ unless otherwise stated.
- **Level vs lattice.** We write $\ell \in \mathbb{N}$ for *level* and $l \in \mathbb{N}$ for the *lattice* parameter to avoid clashes.

Operators

Let d denote the raw integer digit arrays $\{d_{\ell,t,r}\}$. We define:

$$\begin{aligned}
&\text{Tick at } (\ell^*, r^*, t^*): && T_{\ell^*,r^*,t^*}(d) \;=\; d + e_{\ell^*,r^*,t^*}, && (3)\\
&\text{Normalization (per track):} && N(d) \;=\; \text{deterministic tri-site normalization to } \mathcal{D}, && (4)\\
&\text{Möbius holonomy:} && R(d) \text{ acts by } t \mapsto -t \text{ on chosen tracks}, && (5)\\
&\text{Circular/spiral shift:} && S_\kappa(d) \text{ acts by } t \mapsto t + \kappa,\ \kappa \in \mathbb{Z}. && (6)
\end{aligned}$$

A *macrocycle* with N ticks and holonomy $\chi \in \{-1, +1\}$ updates (cf. (14))

$$\mathcal{H}^{(N)}_{\chi,\kappa}(d) \;=\; \begin{cases} R \circ N \circ T^N(d), & \chi = -1 \quad \text{(Möbius)},\\ S_\kappa \circ N \circ T^N(d), & \chi = +1 \quad \text{(circular/spiral)}. \end{cases}$$

Invariants and Lyapunov

- *Value preservation under N.* N preserves $\mathrm{Val}(d) = \sum_{\ell,r,t} s^\ell d_{\ell,t,r} B^{r+lt}$.
- *Lyapunov potential.* Fix $q_\star = \sqrt{\beta_l}$ (so $1 < q_\star < \beta_l$). Define

$$E_{q_\star}(d) \;=\; \sum_{\ell,r,t} s^\ell \, |d_{\ell,t,r}| \, q_\star^t. \tag{7}$$

  Each unit carry strictly reduces $E_{q_\star}$; each tick increases it by at least $s^{\ell^*} q_\star^{t^*}$ before normalization.

- *Center-of-mass on the anchor track.* With $Q^l = +1$, ticks at $(\ell^*, r^*, 0)$ and Möbius holonomy imply $C_{\ell^*,r^*}(d) = \sum_t t \, d_{\ell^*,t,r^*} = 0$ at the end of every Möbius macrocycle.

Quench rule and mode selection

Fix a *collection radius* $t_c \in l\mathbb{Z}_{\geq 1}$ (measured in track sites) and set

$$q_\star := \sqrt{\beta_l}, \qquad E_c := s^{\ell^*} \, q_\star^{t_c}. \tag{8}$$

At the end of a Möbius macrocycle, if $E_{q_\star}(d) \geq E_c$, flip the holonomy to $\chi \leftarrow +1$ for the next macrocycle (*quench*). Post-quench, choose the mode deterministically:

$$\kappa = \begin{cases} 0, & \text{standing (circular) mode,} \\ 1, & \text{running (spiral) mode.} \end{cases} \tag{9}$$

Angular advance and turns. The angular advance per macrocycle is $\Delta\theta = 2\pi\kappa/l$, so the number of turns accrued after n spiral macrocycles is $n\kappa/l$.
Radial scale. Along a fixed track, advancing by l sites multiplies the radius by $\beta_l = b^l$. If the occupied support reaches $t_{\max}$ on any track, the envelope radius satisfies

$$c_1 \, \beta_l^{\lfloor t_{\max}/l \rfloor} \;\leq\; r_{\mathrm{env}} \;\leq\; c_2 \, \beta_l^{\lfloor t_{\max}/l \rfloor}, \qquad c_1 \in [1, b], \; c_2 \in [b^{l-1}, b^l]. \tag{10}$$

Since $E_{q_\star}(d) \geq s^{\ell^*} q_\star^{t_{\max}}$, the quench threshold (8) corresponds to a minimum $t_{\max} \geq t_c$ and hence a minimum radius factor $\beta_l^{t_c/l}$.

Spawn law (toy) and sufficient termination

At the end of each spiral (or circular) macrocycle, for each level ℓ perform an independent Bernoulli trial with probability

$$p_\ell \;=\; p_0 \, \rho^\ell, \qquad 0 < p_0 < 1, \;\; 0 < \rho < 1. \tag{11}$$

On success at level ℓ, add a fixed *child template* $\mathsf{G}_{\ell \to \ell+1}$ with at most L occupied sites per track and digit magnitudes bounded by D; then normalize locally.

Proposition 4.1 (Finite spawns and termination per macrocycle)**.** *If $\{p_\ell\}$ are independent and satisfy $\sum_{\ell \geq 0} p_\ell < \infty$ (true for (11)), then with probability 1 only finitely many levels spawn in any given macrocycle (Borel–Cantelli). Since each spawn adds finite support and normalization terminates trackwise, every macrocycle terminates almost surely.*

Re-closure rule (deterministic schedule)

To guarantee reproducibility, cap the spiral stage to a fixed number $M_{\text{run}} \in \mathbb{N}$ of macrocycles; then set $\chi \leftarrow -1$ (Möbius) for the next macrocycle, regardless of $E_{q_\star}$. This yields the cycle

$$\text{Möbius collection} \xRightarrow{E_{q_\star} \geq E_c} \text{quench to } \chi = +1 \xRightarrow{M_{\text{run}} \text{ macrocycles}} \text{re-closure to } \chi = -1.$$

Remark 4.2. One may supplement this with an energy rule: if during the spiral stage a macrocycle has *no* spawns and uses *no* ticks ($N = 0$), then $E_{q_\star}$ is unchanged and re-closure can be triggered by an *energy window* $E_{q_\star} \in [E_{\min}, E_{\max}]$ chosen a priori. The deterministic cap M_{run} remains the reproducibility guarantee.

Worked example ($\varphi, l = 4$) with a short tick trace

Here $b = \varphi = \frac{1+\sqrt{5}}{2}$, $l = 4$, so $\beta_4 = \varphi^4 \approx 6.854$, $q_\star = \sqrt{\beta_4} = \varphi^2 \approx 2.618$, and $V_4 = L_4 = 7$. Use $\mathcal{D} = \{0, 1, \ldots, 6\}$ and ticks at $(\ell^*, r^*, t^*) = (0, 0, 0)$ during Möbius collection.

Tick-by-tick on the anchor track (normalized state shown). Let (d_{-1}, d_0, d_{+1}) denote digits at $t = -1, 0, +1$ on residue $r = 0$.

tick #	state (d_{-1}, d_0, d_{+1})	$E_{q_\star}$
0	$(0, 0, 0)$	0
1	$(0, 1, 0)$	$q_\star^0 = 1$
2	$(0, 2, 0)$	2
$\vdots$	$\vdots$	$\vdots$
6	$(0, 6, 0)$	6
7	*carry at* t=0 $\Rightarrow (1, 0, 1)$	$q_\star^{-1} + q_\star^{+1} \approx 0.382 + 2.618 = 3.000$
8	$(1, 1, 1)$	$q_\star^{-1} + 1 + q_\star^{+1} \approx 0.382 + 1 + 2.618 = 4.000$
$\vdots$	$\vdots$	$\vdots$
13	$(1, 6, 1)$	$q_\star^{-1} + 6 + q_\star^{+1} \approx 0.382 + 6 + 2.618 = 9.000$
14	*carry at* t=0 $\Rightarrow (2, 0, 2)$	$2\,(q_\star^{-1} + q_\star^{+1}) \approx 2 \times 3.000 = 6.000$

This illustrates: (i) symmetry around $t = 0$, (ii) outward growth of support, and (iii) the Lyapunov behavior (carries reduce $E_{q_\star}$ locally).

Quench parameters. Choose $t_c = 8$ (two full turns on a track). Then $E_c = s^0\, q_\star^8 = (\varphi^2)^8 = \varphi^{16}$. Upon hitting this threshold at the end of a Möbius macrocycle, flip to $\chi = +1$ and choose $\kappa = 1$. Set a reproducible cap $M_{\text{run}} = 5$ spiral macrocycles before re-closure.

Spawn law. Pick $p_0 = 0.2$ and $\rho = \frac{1}{3}$ in (11), with a child template bounded by $L = 3$ sites per track and $D = 6$ digits. Then $\sum_\ell p_\ell = \frac{p_0}{1-\rho} = 0.3 < \infty$, so macrocycles terminate almost surely.

Cross-reference

These rules rely on the phase-aligned identity (2) and the normalization theorem (termination and confluence) proved in the math draft [1].

5 Temporal envelopes and ticks

We model the temporal memory as a hierarchical register with an *envelope type* (holonomy) and a *tick* mechanism.

5.1 State

For each level $\ell \geq 0$ and residue $r \in \{0, \ldots, l-1\}$ we store digits $d_{\ell,t,r} \in \mathbb{Z}$ at sites $t \in \mathbb{Z}$. The register's value is

$$Z \;=\; \sum_{\ell \geq 0} s^{\ell} \sum_{r=0}^{l-1} \sum_{t \in \mathbb{Z}} d_{\ell,t,r}\, B^{\,r+lt}. \tag{12}$$

The envelope holonomy is the discrete variable $\chi \in \{-1, +1\}$: $\chi = -1$ (Möbius) denotes orientation-reversing monodromy per loop; $\chi = +1$ (circular/spiral) is orientation-preserving.

5.2 Tick operator and normalization

A *tick* at site (ℓ^*, r^*, t^*) adds $+1$ then normalizes per track:

$$\mathsf{T}_{\ell^*,r^*,t^*} : \quad d \;\mapsto\; \mathcal{N}\big(d + e_{\ell^*,r^*,t^*}\big), \tag{13}$$

where $\mathcal{N}$ is the deterministic tri-site normalization and $e_{\ell,r,t}$ is the basis vector at that site. Define the Lyapunov $E_q(d) = \sum_{\ell,r,t} s^{\ell} |d_{\ell,t,r}| q^t,\ q \in (1, \beta_l)$.

Lemma 5.1 (Tick and carry monotonicity)**.** *Each tick increases E_q by at least $s^{\ell^*} q^{t^*}$ before normalization; each carry strictly reduces E_q.*

5.3 Macrocycle and holonomy action

A *macrocycle* applies N ticks, then the holonomy action. For Möbius we use index reflection $\mathsf{R} : t \mapsto -t$ on each affected track; for circular/spiral we use index shift $\mathsf{S}_\kappa : t \mapsto t + \kappa$ with integer κ (pitch). Thus

$$d^+ \;=\; \begin{cases} \mathsf{R} \circ \mathcal{N} \circ \mathsf{T}^N(d), & \chi = -1 \quad \text{(Möbius)}, \\ \mathsf{S}_\kappa \circ \mathcal{N} \circ \mathsf{T}^N(d), & \chi = +1 \quad \text{(circular/spiral)}. \end{cases} \tag{14}$$

Because overlapping carries commute on each track, the order of the N ticks inside a macrocycle is irrelevant.

6 Prediction: Chebyshev–Lucas Radial Law

Theorem 6.1 (Exact radial decomposition)**.** *Assume $Q^l = +1$. Let $d_t(T)$ be the normalized digits on the anchor track after T ticks at $t = 0$. Then $d_{-t}(T) = d_t(T)$ and*

$$T \;=\; d_0(T) \;+\; \sum_{j \geq 1} d_j(T)\, U_j(V_l),$$

where $U_0(V) = 2$, $U_1(V) = V$, $U_{j+1}(V) = V U_j(V) - U_{j-1}(V)$ (i.e. $U_j(V) = 2T_j(V/2)$ with T_j Chebyshev polynomials of the first kind).

Proof. Let $r > 1$ solve $r^2 - V_l r + 1 = 0$. Define $I(d) = \sum_t d_t r^t$. A tick at $t = 0$ adds $+1$ to I. A unit carry at site t changes I by $-V_l r^t + r^{t+1} + r^{t-1} = 0$. Hence starting from $I = 0$ we get $I(d(T)) = T$. By symmetry $d_{-t} = d_t$,

$$T \;=\; d_0 + \sum_{j \geq 1} d_j\, (r^j + r^{-j}),$$

and since $r + r^{-1} = V_l$, the quantity $r^j + r^{-j}$ is the polynomial $U_j(V_l)$ defined by $U_0 = 2$, $U_1 = V_l$, $U_{j+1} = V_l U_j - U_{j-1}$. □

Remark 6.2. This gives a unique, palindromic expansion of T in the Chebyshev–Lucas basis $\{U_j(V_l)\}$ with digits $0 \leq d_j \leq V_l - 1$ determined by normalization.

7 Stage I (Möbius): collection without drift

Assume ticks arrive at a fixed anchor $(\ell^*, r^*, t^* = 0)$ each macrocycle while $\chi = -1$.

Proposition 7.1 (Collection and symmetry)**.** *Suppose $Q^l = +1$ (e.g., $b = \varphi$, even l). Under the Möbius update (14), the track (ℓ^*, r^*) remains mirror-symmetric in t, its occupied support radius grows monotonically with the cumulative tick count, and its index center of mass vanishes for all macrocycles: $\sum_t t\, d_{\ell^*,t,r^*} = 0$.*

Sketch. Reflection R and a fixed injection site preserve symmetry. Tri-site carries send $(t) \mapsto (t-1, t, t+1)$ with neighbor increments $\pm Q^l$; for $Q^l = +1$ both neighbors increase together, pushing support outward while keeping mean zero. Termination per macrocycle follows from the Lyapunov argument. □

8 Quench: unwrap to circular or spiral

We impose a threshold rule that flips holonomy when the register is "full".

Definition 8.1 (Quench)**.** Fix a threshold $E_c > 0$. When $E_q(d) \geq E_c$ at the end of a macrocycle with $\chi = -1$, set $\chi \leftarrow +1$ for the next macrocycle.

Remark 8.2 (Macrocycle monodromy viewpoint). Equivalently, partition a macrocycle into blocks $j = 1, \ldots, J$; each contributes a scale $S_j > 0$, a step count $m_j \in \mathbb{N}$ on lattice (b_j, l_j), and a phase gate θ_j. The total transform is $T = \prod_j S_j b_j^{m_j} \exp\big(i\, 2\pi \sum_j m_j/l_j\big) e^{i \sum_j \theta_j}$. A "half-twist" condition (monodromy -1) is $\prod_j S_j b_j^{m_j} = 1$, $\sum_j m_j/l_j + \frac{1}{2\pi}\sum_j \theta_j \equiv \frac{1}{2} \pmod 1$. Quench corresponds to switching from $T = -1$ to $T = +1$.

9 Stage II (circular/spiral): propagation and fractal children

With $\chi = +1$, choose an integer *pitch* κ. The macrocycle update is $d^+ = \mathsf{S}_\kappa \circ \mathcal{N} \circ \mathsf{T}^N(d)$.

Proposition 9.1 (Standing vs. running modes)**.** *If $\kappa = 0$ (circular) there is no net index drift on the anchor track; if $\kappa \neq 0$ (spiral) there is a drift of κ sites per macrocycle. The tri-site normalization and its bounds are invariant under S_κ.*

9.1 Fractal spawn across levels

Let $\Theta_\ell > 0$ be level thresholds. When $E_q(d_{\ell,\cdot,\cdot}) \geq \Theta_\ell$, spawn a child pattern at level $\ell + 1$:

$$d_{\ell+1,\cdot,\cdot} \;\leftarrow\; d_{\ell+1,\cdot,\cdot} \;+\; \mathsf{G}_{\ell\to\ell+1}, \tag{15}$$

where $\mathsf{G}_{\ell\to\ell+1}$ is a finite-support template (e.g., a small loop on each residue) that will be scaled by s in the value. Normalization then runs as usual.

Theorem 9.2 (Convergence and normalizability under spawn)**.** *If $s < 1$ and each $\mathsf{G}_{\ell\to\ell+1}$ has finite support per track, then for any finite number of spawns per macrocycle, the value Z converges absolutely and the post-spawn state is normalizable (terminates) at every macrocycle.*

Sketch. Your convergence estimate gives $\sum_\ell s^\ell C_\ell < \infty$ when each level has finite support; normalization terminates by the Lyapunov decrease of carries. □

10 Termination and re-closure to Möbius

When outward propagation has relaxed, we re-close the envelope.

Definition 10.1 (Re-closure rule)**.** Fix $E_{\text{close}} > 0$ with $E_{\text{close}} < E_c$. If, at the end of a circular/spiral macrocycle, $E_q(d) < E_{\text{close}}$ and no level satisfies $E_q(d_\ell) \geq \Theta_\ell$, set $\chi \leftarrow -1$ for the next macrocycle.

Remark 10.2. This yields the full cycle: *Möbius collection* $\Rightarrow$ *quench* $\Rightarrow$ *circular/spiral with fractal spawn* $\Rightarrow$ *re-closure to Möbius.*

11 Algorithms (macrocycle skeleton)

Input: state d, holonomy χ, thresholds $(E_c, E_{\text{close}}, \{\Theta_\ell\})$, pitch κ, ticks per macrocycle N.
Loop over macrocycles:

1. Apply N ticks at $(\ell^*, r^*, 0)$ and normalize per track.
2. If $\chi = -1$, apply reflection R; if $\chi = +1$, apply shift S_κ.
3. For each level ℓ, if $E_q(d_\ell) \geq \Theta_\ell$, add child template at level $\ell + 1$ and re-normalize locally.
4. Update χ by quench/re-closure rules.

12 Worked example: φ with quarter-turn lattice

Set $b = \varphi = \frac{1+\sqrt{5}}{2}$, $l = 4$ so $Q^l = +1$ and $V_l = L_4 = 7$. Use nonnegative digits $\{0, \ldots, 6\}$.

- **Möbius collection:** ticks at $(\ell^* = 0, r^* = 0, t = 0)$; the track remains symmetric and fills outward.
- **Quench:** when E_q crosses E_c, set $\chi \leftarrow +1$.
- **Spiral phase:** choose $\kappa = 1$ to get a helix in the time-circle embedding; set $s = \varphi^{-4}$ for level self-similarity after one full turn.
- **Re-closure:** when the register relaxes below E_{close} and no level triggers spawn, set $\chi \leftarrow -1$.

13 Discussion and outlook

The (s, b, l) calculus provides a discrete, first-principles engine for temporal envelopes. The Möbius $\leftrightarrow$ circular/spiral transition is captured by a holonomy flip driven by a Lyapunov threshold. The child spawn mechanism leverages the hierarchy to produce fractal bouquets while preserving the normalization guarantees. Connections to continuous models include (i) helical visualization with time drawn as a circle, where pitch is proportional to the index drift κ, and (ii) Kaluza–Klein optics where null rays project to geodesics of an optical metric (not developed here).

Acknowledgments

We thank prior work on complex bases and normalization; our phase-aware, tri-site calculus and its dynamics build directly on those ideas.

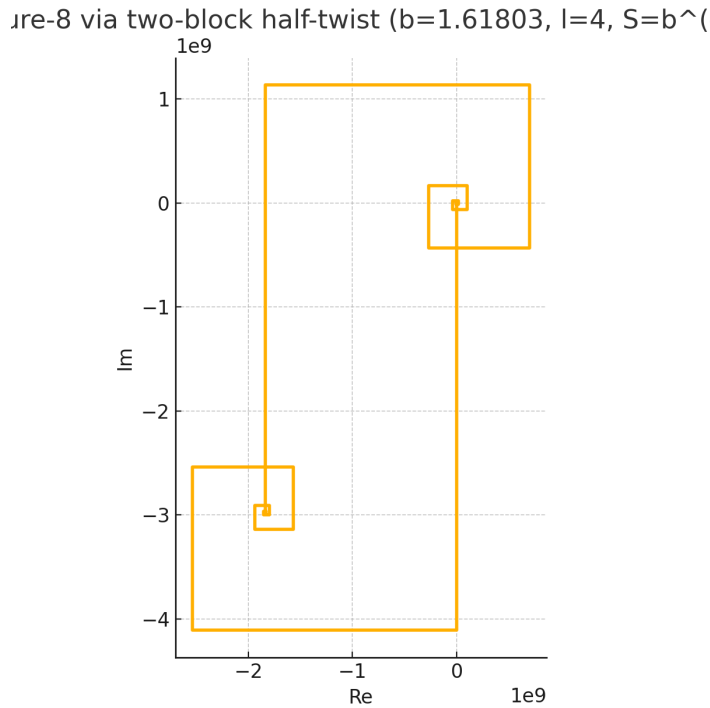

Figure 1: **Half-twist macrocycle (lemniscate-like)** for $(b, l) = (\varphi, 4)$ using two equal blocks with a π rotation and per-block scale $S = b^{-1}$, yielding monodromy -1 and a closed figure-8 centerline.

References

[1] V. R. and A. I. Local Normal Forms for Complex Base Expansions: A General (s, b, l) Framework and the φ Specialization. (2025).

[2] D. E. Knuth. The Art of Computer Programming, Vol. 2: Seminumerical Algorithms (3rd ed.). Addison–Wesley, 1997.

[3] W. Penney. A "binary" system for complex numbers. *J. ACM* 12(2):247–248, 1965.

[4] W. J. Gilbert. Radix representations of quadratic fields. *J. Math. Anal. Appl.* 83(1):264–274, 1981.

[5] W. J. Gilbert. Complex bases and Rademacher functions. *J. Math. Anal. Appl.* 94(2):438–454, 1983.

[6] C. Frougny and J. Sakarovitch. Number representation and finite automata. In *Combinatorics, Automata and Number Theory*, 1992.

[7] C. Heuberger. Canonical number systems in imaginary quadratic fields. *Math. Comp.* 76(257):195–210, 2007.

[8] S. Akiyama and A. Pethő. Canonical number systems: a survey. In *Geometry and Analysis of Fractals*, Springer, 2011.

14 Bridge to Curvature

Let $g = -\alpha(x)^2(d\theta + A)^2 + h(x)$ on $\Sigma \times S^1_\theta$.

Proposition 14.1 (Optical metric law)**.** *Future-directed null geodesics of g project to geodesics of the optical metric $\hat{h} = \alpha^{-2}h$ on Σ.*

Remark 14.2 (Curvature triggers). Nonconstant α and nonzero curvature $F = dA$ of the temporal connection produce nonzero curvature in g (via KK reduction). Hence hierarchical (s, b, l) fields (α, A) naturally induce curvature where child envelopes spawn.